ANALYZING E-LEARNING PLATFORM PURCHASES

1. Project Overview and Objective

- **Overview**: This project focuses on the design and implementation of a relational database for an e-learning platform. Schema created through structuring data into three primary entities - Learners, Courses, and Purchases. The project demonstrates how SQL can be used to manage and track digital transactions across a global user base.
- **Objective**: The primary goal is to transform raw purchase data into actionable business intelligence. This involves setting up a robust database schema, ensuring data integrity through primary and foreign keys, and performing advanced data analysis to evaluate sales performance and learner engagement.

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2. Data Sources

- **Source Description and Timeline:** Manually created E-learning platform data for the Months November 2025, December 2025, January 2026, February 2026, March 2026.

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3. Problem Statement

- **The Problem**: Currently, the e-learning platform's data is fragmented, making it difficult to identify which courses are performing well and which regions are driving the most revenue. Without a centralized relational database, we cannot easily track individual learner's behaviors or identify trends across different course categories (e.g., Programming vs. Design).
- **The Goal**: The goal is to develop a MySQL-driven solution that provides a "single source of truth" for the platform. By utilizing SQL joins and aggregations, the project aims to answer critical business questions, such as identifying top-selling courses, calculating total revenue by category, and pinpointing countries with the highest number of active learners.

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4. Attribute (Column/Features) Details

Attribute Name	Data Type	Description
learner_id	Integer	Primary key indicating learners' unique enrolment
full_name	Varchar	Learner's names
Country	Varchar	Place from where Learners purchased the course

course_id	Varchar	Unique ID for the course
course_name	Varchar	Names of the e-learning courses
Category	Varchar	Identification of courses according to the department
learner_id	Integer	Unique ID for the learners
Quantity	Integer	Estimate of course bought
unit_price	Decimal	Cost of course
purchase_date	Date	Date course was bought
Purchase_ID	Unique	Unique ID for purchase

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5. Tools & Technologies

- **SQL:** Creating database and schema. Data population. Data exploration.

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6. Key Insights

- **Total Spending by Learner & Country** :

Result Grid Filter Rows:

full_name	country	total_spent
Charlotte Anderson	Australia	1,099.98
Seethai Lakshmi	India	1,099.98
Alaric Sterling	UK	1,099.98
Mary Smith	USA	1,099.98
Udhaya Nidhi	India	1,099.98
Robert Evans	USA	1,099.98
Hema Parichelvan	India	1,099.98
Henry Johnson	Australia	1,099.98
Mia White	Australia	1,099.98
Poorna Chandran	India	639.98
Citlalli Andazola	Mexico	639.98
Saroja Pooranacha...	India	639.98
Noah Roy	Canada	639.98
Jean Richard	France	639.98
Shivani Iyer	India	599.99
Sundar Pichai	India	599.99
Sathya Moorthy	India	599.99
Maximillan Richter	Germany	599.99
Emma Brown	UK	599.99
Paloma Noh	Mexico	599.99
Evelyn Garcia	Mexico	599.99
Consuelo Zarate	Mexico	599.99
Karthik Muralidharan	India	599.99
Alex Vondra	Australia	599.99

639.98

- **Countries buying more Courses**: Even though Mexico is on the top showing more number of courses being purchased, India being second has generated more revenue.

Result Grid | Filter Rows: | Expo

country	total_courses_purchased	total_revenue
Mexico	30	4909.70
India	27	10149.73
Germany	19	2679.81
Australia	16	4739.84
UK	12	3009.88
Canada	12	3679.88
France	10	1079.90
USA	10	3399.90

- **Total Revenue**: Total revenue generated from November 2025 through March 2026 is 33648.64.

Result Grid | Filter Rows:

Grand_Total_Revenue
33648.64

- **Total Revenue Generated per Course**: The data revels Python Mastery, Tableau/PowerBI, MySQL and UX/UI & Digital Design are selling highly.

course_name	category	Total_Revenue	total_quantity_sold
UX/UI & Digital Design	Design	9599.84	16
Graphic Design Basics	Design	7499.85	15
Java Development	Programming	7199.88	12
Data Science Essentials	Data Science	2999.85	15
Digital Marketing 101	Marketing	2399.88	12
Tableau/Power BI	Data Science	1359.83	17
MySQL for Analytics	Data Science	1119.84	16
Marketing Analytics Foundation	Marketing	749.85	15
Python Mastery	Programming	719.82	18

- **Revenue by Category**: The top revenue generating category is Design.

Result Grid | Filter Rows:

category	total_orders	total_revenue
Design	31	17,099.69
Programming	30	7,919.70
Data Science	48	5,479.52
Marketing	27	3,149.73

- **Total Monthly Revenue**: November and December month generated the top most revenue.

Result Grid | Filter Rows: | Export:

Year	Month	Monthly_Revenue	Total_Transactions
2025	November	12159.64	36
2025	December	6639.71	29
2026	January	6509.74	26
2026	February	2729.82	18
2026	March	5609.73	27

- **TOP 3 Most Purchased Course**: Python Mastery, Tableau/Power BI and MySQL for Analytics are the most purchased courses.

Result Grid | Filter Rows:

course_name	total_quantity_sold
Python Mastery	18
Tableau/Power BI	17
MySQL for Analytics	16

- **Category Revenue & Unique Learners**: Unique learners are more in programming category.

Result Grid | Filter Rows: | Exp

category	total_revenue	unique_learners
Data Science	5,479.52	24
Design	17,099.69	22
Marketing	3,149.73	20
Programming	7,919.70	25

- **Learners with Multi-Category Purchase**: Some of the learners have purchased related courses from different categories.

Result Grid | Filter Rows:

full_name	categories_purchased
Adelle Velasco	2
Deepak Malhotra	2
Felix Hoffmann	2
Juan Martinez Gonzalez	2
Noah Wilson	2

- **Average Courses per Transaction**: 1.

Result Grid | Filter Rows:

average_courses_per_order
1.0000

- **Best Selling Bundles**: Data Science Essentials and MySQL for Analytics have been the top selling combination bundle.

Result Grid | Filter Rows:

Course_A	Course_B	Times_Bought_Together
DA-2025	MY-2025	11
GR-2025	UX-2025	9
MY-2025	TA-2025	9
DA-2025	TA-2025	9
JA-2025	PY-2025	5

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

full_name	courses_purchased
Abgail Robinson	Data Science Essentials, MySQL for Analytics, Tableau/Power BI
Adelle Velasco	Data Science Essentials, Tableau/Power BI, MySQL for Analytics, Python Mastery
Agustin Yam	MySQL for Analytics, Data Science Essentials, Tableau/Power BI
Alaric Sterling	UX/UI & Digital Design, Graphic Design Basics
Amelia Martin	Digital Marketing 101, Marketing Analytics Foundation
Anna Schmidt	Digital Marketing 101, Marketing Analytics Foundation
Beatrix Fallow	Data Science Essentials, Tableau/Power BI, MySQL for Analytics
Benjamin Mueller	Data Science Essentials, MySQL for Analytics
Charlotte Anderson	UX/UI & Digital Design, Graphic Design Basics
Citlalli Andazola	Python Mastery, Java Development
Deepak Malhotra	Data Science Essentials, Python Mastery, Tableau/Power BI, MySQL for Analytics
Diego Torres	Data Science Essentials, Tableau/Power BI, MySQL for Analytics
Felix Hoffmann	MySQL for Analytics, Data Science Essentials, Python Mastery, Tableau/Power BI
Guadalupe Rodriguez	MySQL for Analytics, Tableau/Power BI
Hawthorn Dubois	MySQL for Analytics, Tableau/Power BI
Hema Parichelvan	UX/UI & Digital Design, Graphic Design Basics
Henry Johnson	UX/UI & Digital Design, Graphic Design Basics
Isla Jones	Digital Marketing 101, Marketing Analytics Foundation
Jean Richard	Python Mastery, Java Development
Juan Martinez Gon...	Tableau/Power BI, Tableau/Power BI, Data Science Essentials, Python Mastery
Liam Smith	Marketing Analytics Foundation, Digital Marketing 101
Lucia Cruz Morales	Tableau/Power BI, MySQL for Analytics, Data Science Essentials
Lukas Muller	MySQL for Analytics, Data Science Essentials
Mary Smith	Graphic Design Basics, UX/UI & Digital Design

- **Unpurchased Courses**: Looking at the data, there are no unpurchased courses currently.

Result Grid | Filter Rows:

course_name	category

- **Monthly Sales Trend**: Sales were night in the month of November.

Result Grid | Filter Rows:

purchase_month	monthly_revenue
2025-11	12,159.64
2025-12	6,639.71
2026-01	6,509.74
2026-02	2,729.82
2026-03	5,609.73

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7. Conclusions

- Programming and Data Science categories are likely the highest revenue drivers based on the quantity sold.
- A small percentage of learners are purchasing from multiple categories, suggesting an opportunity for cross-selling.
- Focus marketing efforts on the "Top 3" courses identified to maximize ROI.
- Consider discounting or re-evaluating courses that are selling low.